Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-8067
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

The template focuses on larger user bases, increased data loads, extended features and a future roadmap. For this ServiceNow ACL project, the plan extends the current role- and Branch-based security model.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Access rules currently focus on one custom table and Branch values ECE/EEE/CSE.	Additional departments or tables require additional ACL configuration.	Medium
2	CRUD permissions are represented by separate custom roles bb1–bb4.	Larger role models may become harder to maintain manually.	Medium
3	Verification is based on manual impersonation and list/form checks.	Large changes require more systematic regression testing.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Custom roles and users are managed in ServiceNow.	Use structured role/group administration and standardize access assignment for larger user populations.
2	Data Storage	Records are stored in the u_institution_details custom table.	Extend the data model with additional fields/tables only when required; maintain appropriate ACLs for each new data area.
3	Performance	ACL checks are evaluated by the ServiceNow platform.	Measure response time and query behavior after significant data growth; review ACL conditions and list configurations if needed.
4	Security	READ uses role + Branch condition; CRUD uses operation-specific roles.	Expand to department/branch groups, standardize role hierarchy and add regression/security test cases for every new rule.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Extend field-based restrictions to additional departments or business units.	Next iteration	Broader record-level security coverage.
Phase 2	Standardize role/group-based access administration.	Future sprint	Simpler permission maintenance as users increase.
Phase 3	Automated regression tests for ACL scenarios.	Future sprint	More consistent verification after configuration changes.
Phase 4	Security monitoring and audit-oriented reporting.	Long-term	Improved visibility into access decisions and changes.